

Lab: Creating Your Invention's Documentation System

Objective

Build a documentation system for your invention project that captures your generative model as an external representation — including design decisions, version history, test results, and uncertainties — so that another competent agent could continue your work.

Prerequisites

- Familiarity with lab notebooks, version control, decision records, and uncertainty communication from this module's lecture
- Your invention project materials (prototype, test data, design notes from prior modules)
- A documentation medium (physical notebook, digital document, or wiki)

Materials

- Lab notebook or digital equivalent
- Camera (phone camera is fine) for prototype photography
- Your prototype (or detailed drawings/descriptions)
- All test records from prior modules

Part 1: Lab Notebook Entry (15 minutes)

Goal: Practice writing a lab notebook entry that records one turn of the Active Inference loop in real time.

Write a lab notebook entry for your most recent prototyping or testing activity. Follow these conventions:

Date and time:

Write your response here

Prototype version:

Write your response here

Objective of this session (what hypothesis are you testing or what action are you taking?):

Write your response here

Raw observations (what you saw, measured, or heard — no interpretation yet):

Write your response here

Predictions vs. outcomes:

What you predicted	What you observed	Prediction error

Write your response here

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Write your response here

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Write your response here

Write your response here

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Write your response here

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Write your response here

Write your response here

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Write your response here

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Write your response here

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Interpretation (what you think the observations mean — clearly separated from the raw data):

Write your response here

Decision (what you will do next, and why):

Write your response here

Open questions (what remains uncertain):

Write your response here

Now review your entry: could another person who has never seen your invention read this entry and understand what happened, why it matters, and what you plan to do next?

Write your response here

Part 2: Version History (10 minutes)

Goal: Create a version history record for your prototype, from the earliest concept to the current state.

Document every significant version of your prototype:

Version	Date	Key Changes from Previous Version	Why These Changes Were Made	Result (better/worse/different?)
v0 (concept)				

Write your response here

| **Initial concept** | **Starting point** | **N/A** | |

Write your response here

|

Write your response here

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Write your response here

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Write your response here

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Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

|| Current |

Write your response here

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Write your response here

|

Write your response here

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Write your response here

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Photography protocol: Have you photographed each version? If not, photograph the current version now from at least three angles (front, side, top or detail view).

Describe your current version control practices. What do you do well? What could you improve?

Write your response here

Part 3: Design Decision Record (15 minutes)

Goal: Write a formal design decision record for the most significant decision in your invention project.

Choose the single most important design decision you have made (material choice, mechanism selection, form factor, target user, etc.) and document it:

Decision title:

Write your response here

Decision date:

Write your response here

Context (what situation prompted this decision?):

Write your response here

Decision (what was chosen):

Write your response here

Alternatives considered:

Alternative	Pros	Cons	Why rejected

Write your response here

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Write your response here

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Write your response here

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Write your response here

Write your response here

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Write your response here

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Write your response here

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Write your response here

Write your response here

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Write your response here

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Write your response here

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Write your response here

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Evidence supporting the decision:

Write your response here

Uncertainties remaining:

Write your response here

Reversibility (how hard would it be to change this decision later?):

Write your response here

Consequences (what downstream decisions does this constrain?):

Write your response here

Part 4: Uncertainty Map (10 minutes)

Goal: Create an honest map of what you know, what you think you know, and what you do not know about your invention.

For each major aspect of your invention, assess your confidence level:

Aspect	Current Belief	Confidence Level	Evidence Basis	What would change your belief?
Core mechanism works				

Write your response here

| Tested and confirmed / Estimated / Assumed / Highly uncertain |

Write your response here

|

Write your response here

|| Target user wants this |

Write your response here

| Tested and confirmed / Estimated / Assumed / Highly uncertain |

Write your response here

|

Write your response here

| | Materials are appropriate |

Write your response here

| Tested and confirmed / Estimated / Assumed / Highly uncertain |

Write your response here

|

Write your response here

| | Can be manufactured at scale |

Write your response here

| Tested and confirmed / Estimated / Assumed / Highly uncertain |

Write your response here

|

Write your response here

|| Price point is viable |

Write your response here

| Tested and confirmed / Estimated / Assumed / Highly uncertain |

Write your response here

|

Write your response here

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What is the single biggest "assumed but untested" belief in your project?

Write your response here

Part 5: The Reconstruction Test (10 minutes)

Goal: Test whether your documentation could enable another person to understand your invention and continue your work.

Imagine you are handing your entire project to a collaborator who has never seen it. They receive only your documentation (lab notebook entries, version history, decision records, test data, and uncertainty map).

Could they answer these questions from your documentation alone?

Question	Could they answer it? (Yes/Partially/No)	What's missing?
What does the invention do?		

Write your response here

|

Write your response here

| | **How does it work?** |

Write your response here

|

Write your response here

| | **What has been tested?** |

Write your response here

|

Write your response here

| | What works and what does not? |

Write your response here

|

Write your response here

| | Why were key design decisions made? |

Write your response here

|

Write your response here

| | What is uncertain? |

Write your response here

|

Write your response here

| | What should be done next? |

Write your response here

|

Write your response here

|

For each "No" or "Partially" answer, what documentation would you need to create to fill the gap?

Write your response here

Discussion and Debrief

1. **Documentation debt:** How much of your invention's knowledge currently exists only in your head? Was this exercise a wake-up call about documentation gaps?
2. **Decision amnesia:** Were you able to reconstruct the rationale for your key design decisions? Had you forgotten some of the reasoning? What does this suggest about the value of contemporaneous documentation?
3. **Uncertainty honesty:** Was it uncomfortable to formally acknowledge what you do not know? Why might investors, managers, or collaborators benefit from seeing your uncertainty map?
4. **Version control gaps:** What information about previous prototype versions has been lost? How would having that information change your current work?
5. **The future reader:** If you revisit this project in five years, will your documentation make sense to future-you? What would future-you wish you had recorded?

Write your response here